ImpedanceDiffusion: Diffusion-Based Global Path Planning for UAV Swarm Navigation with Generative Impedance Control

Faryal Batool*, Yasheerah Yaqoot*, Muhammad Ahsan Mustafa, Roohan Ahmed Khan, Aleksey Fedoseev, and Dzmitry Tsetserukou

***Abstract*— Safe swarm navigation in cluttered indoor environment requires long-horizon planning, reactive obstacle avoidance, and adaptive compliance. We propose *ImpedanceDiffusion*, a hierarchical framework that leverages image-conditioned diffusion-based global path planning with Artificial Potential Field (APF) tracking and semantic-aware variable impedance control for aerial drone swarms.**

The diffusion model generates geometric global trajectories directly from RGB images without requiring explicit map construction. These trajectories are tracked by an APF-based reactive layer, while a VLM–RAG module performs semantic obstacle classification with 90% retrieval accuracy to adapt impedance parameters for mixed obstacle environments during execution. Two diffusion planners are evaluated: (i) a top-view long-horizon planner using single-pass inference and (ii) a first-person-view (FPV) short-horizon planner deployed via a two-stage inference pipeline on top view. Both planners achieve a 100% trajectory generation rate across twenty static and dynamic experimental configurations and are validated via zero-shot sim-to-real deployment on Crazyflie 2.1 drones through the hierarchical APF–impedance control stack. The top-view planner produces smoother trajectories that yield conservative tracking speeds of 1.0–1.2 m/s near hard obstacles and 0.6–1.0 m/s near soft obstacles. In contrast, the FPV planner generates trajectories with greater local clearance and typically higher speeds, reaching 1.4–2.0 m/s near hard obstacles and up to 1.6 m/s near soft obstacles. Across 20 experimental configurations (100 total runs), the framework achieved a 92% success rate while maintaining stable impedance-based formation control with bounded oscillations and no in-flight collisions, demonstrating reliable and adaptive swarm navigation in cluttered indoor environments.

Video of ImpedanceDiffusion: https://youtu.be/J-Ec2arIJVw
GitHub of ImpedanceDiffusion: https://github.com/Faryal-Batool/ImpedanceDiffusion

Keywords: Diffusion Planning, Swarm Robotics, Impedance Control, Vision-Language Models, Retrieval-Augmented Generation, Variable Compliance, APF

I. Introduction

Safe autonomous navigation in cluttered indoor environments remains a fundamental challenge for aerial robots. Corridors, warehouses, and laboratories present narrow passages, dynamic obstacles, varying illumination, and frequently lack reliable prior maps. Conventional navigation pipelines typically depend on explicit geometric representations (e.g., occupancy maps) and perception frameworks that fuse multiple sensors and intermediate outputs such as depth, segmentation, and mapping; these pipelines often degrade under limited onboard sensing and partial observability.

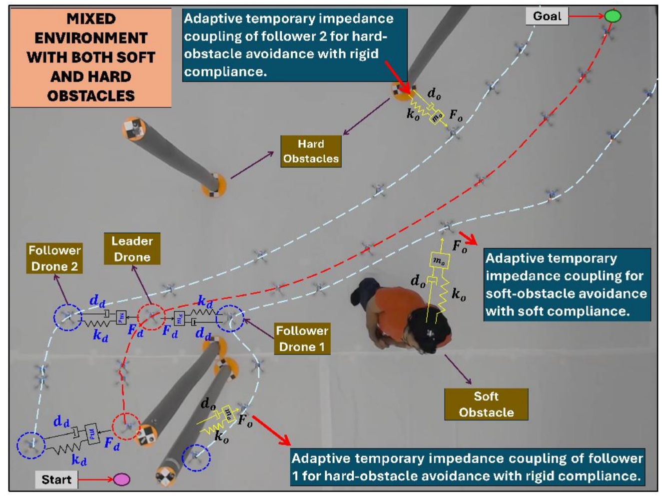

Fig. 1: *ImpedanceDiffusion* in a mixed hard (rigid poles) and soft (human) obstacle environment. Diffusion-based planning generates the global path, APF ensures reactive avoidance (red dashed), and adaptive impedance links maintain formation cohesion and compliant interaction during flight.

This motivates the following question: *Can a drone infer safe global flight trajectories directly from a single RGB image, given only start and goal locations, without explicit maps or hand-crafted heuristics?*

Diffusion models offer a generative alternative for vision-conditioned trajectory synthesis. By learning trajectory distributions from image–trajectory pairs via iterative denoising, the model can implicitly encode traversability cues from raw visual input, enabling smooth and coherent path generation without explicit obstacle or environment maps. However, safe operation in human-shared environments requires more than geometric collision avoidance; it additionally demands adaptive and context-aware compliance.

Our prior work, *ImpedanceGPT* [1], introduced a semantic-driven VLM–RAG framework for impedance adaptation in swarm navigation, where obstacle understanding is used to select impedance parameters for interaction. However, it relied on Artificial Potential Fields (APF) for global planning and predefined impedance ranges per obstacle category. In this work, we propose a hierarchical framework in which an image-conditioned diffusion model generates a *global* path, while APF-based reactive tracking and generative impedance control provide robust avoidance for mixed

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and previously unseen obstacle environments (Fig. 1). We evaluate two diffusion planners with different visual conditioning and planning characteristics: a top-view planner designed for long-horizon global trajectory generation, and a second planner trained on first-person-view (FPV) data for short-horizon planning, which is deployed in a zero-shot manner using a two-stage inference pipeline for global trajectory generation. This enables a comparative analysis of planning behavior under different visual modalities and horizon assumptions. The main contributions of this work are:

- An image-conditioned diffusion planner trained in simulation that generates smooth *global* trajectories directly from a single RGB image and start/goal inputs, without explicit map construction.
- A hierarchical planning–control integration in which classical reactive tracking and adaptive impedance control collectively ensure stable trajectory following and obstacle avoidance of diffusion-generated global paths.
- A semantic-driven VLM–RAG framework that dynamically adapts swarm impedance parameters for mixed soft and hard obstacles based on obstacle understanding.
- A quantitative comparison between top-view and FPV-conditioned diffusion planners, validated through zero-shot sim-to-real deployment on a Crazyflie swarm.

II. Related Work

Safe navigation for aerial robots in indoor environments requires the integration of global planning, local collision avoidance, and adaptive interaction control. While these components have been extensively studied individually, unified frameworks that combine visual trajectory generation, semantic reasoning, and compliant swarm coordination remain less explored within cohesive swarm-level architectures.

A. Diffusion-Based Trajectory Generation

Diffusion models have recently emerged as generative priors for robotic trajectory planning. Methods such as NoMaD, NavDP, and NaviDiffusor learn goal-conditioned diffusion policies that generate action sequences from visual observations, inducing smooth and collision-aware trajectories [2]–[4]. In mapless and long-horizon settings, DTG introduced a conditional diffusion architecture that optimizes travel distance and traversability [5]. Similarly, DiPPeST applied image-conditioned diffusion planning to legged robots, achieving fast trajectory generation and reactive local refinement without classical planners [6].

While diffusion-based planners address geometric feasibility at the global level, they lack mechanisms to regulate physical interaction once robots operate in close proximity to dynamic or heterogeneous obstacles. This limitation motivates the integration of active compliance control with learned global planning.

B. Impedance Control for Aerial and Swarm Systems

Impedance control, introduced by Hogan [7], establishes a force–motion relationship, which enables natural and easily tunable compliant behavior during interaction compared to position or velocity controllers. In aerial robotics, impedance-based approaches have improved trajectory tracking and disturbance handling, including Online Impedance Adaptive Control strategies [8] and aerial manipulation frameworks [9], [10]. Intelligent impedance adaptations have also been explored to enhance robustness [11].

For swarm systems, virtual impedance links have been proposed to achieve safe and coordinated multi-quadrotor behavior [12], [13]. APF combined with impedance control has further improved swarm stability [14]. However, most prior works rely on fixed or manually tuned impedance parameters that remain constant within a given scenario, and cannot explicitly differentiate interaction behavior across heterogeneous obstacle types. As a result, interaction compliance is typically scenario-dependent rather than obstacle-class-dependent.

C. Vision-Language Models for Semantic Navigation

Vision-Language Models (VLMs) provide semantic understanding beyond geometric perception. Recent aerial navigation works such as *See, Point, Fly* demonstrate learning-free VLM-based waypoint grounding [15], while Transformer-based vision-language navigation frameworks enable quadcopter trajectory reasoning through multimodal fusion of visual and language inputs [16]. Other approaches integrate VLM reasoning for socially aware navigation and semantic trajectory selection [17], [18]. Large multimodal models such as Molmo-7B-O, along with embedding methods such as Sentence-BERT further enhance contextual understanding [19], [20]. However, VLM-based reasoning has not been directly coupled with adaptive impedance control for swarm-level compliant interaction.

While previous impedance-based navigation systems employ fixed parameter settings for predefined scenarios, real-world environments frequently contain mixed obstacle types that require differentiated interaction behavior. Building upon [1], which focused on fixed impedance parameters, the present work advances this direction by applying image-conditioned diffusion planning and introducing obstacle-class-dependent impedance adaptation. Compliance parameters are selected based on semantic scene understanding rather than global scenario categorization, allowing different obstacle types to induce distinct interaction behaviors within the same trajectory.

III. ImpedanceDiffusion Framework

The proposed framework extends ImpedanceGPT [1] by employing a diffusion-based global trajectory generator within a hierarchical control architecture. The system comprises four modules: (1) an image-conditioned diffusion model for global path planning, (2) an Artificial Potential Field (APF) as a local planner for tracking by the leader drone and reactive avoidance, (3) a VLM–RAG module for semantic scene understanding and impedance selection, and (4) a virtual impedance-based formation controller for follower drones and drone–obstacle interaction as illustrated in Fig. 2.

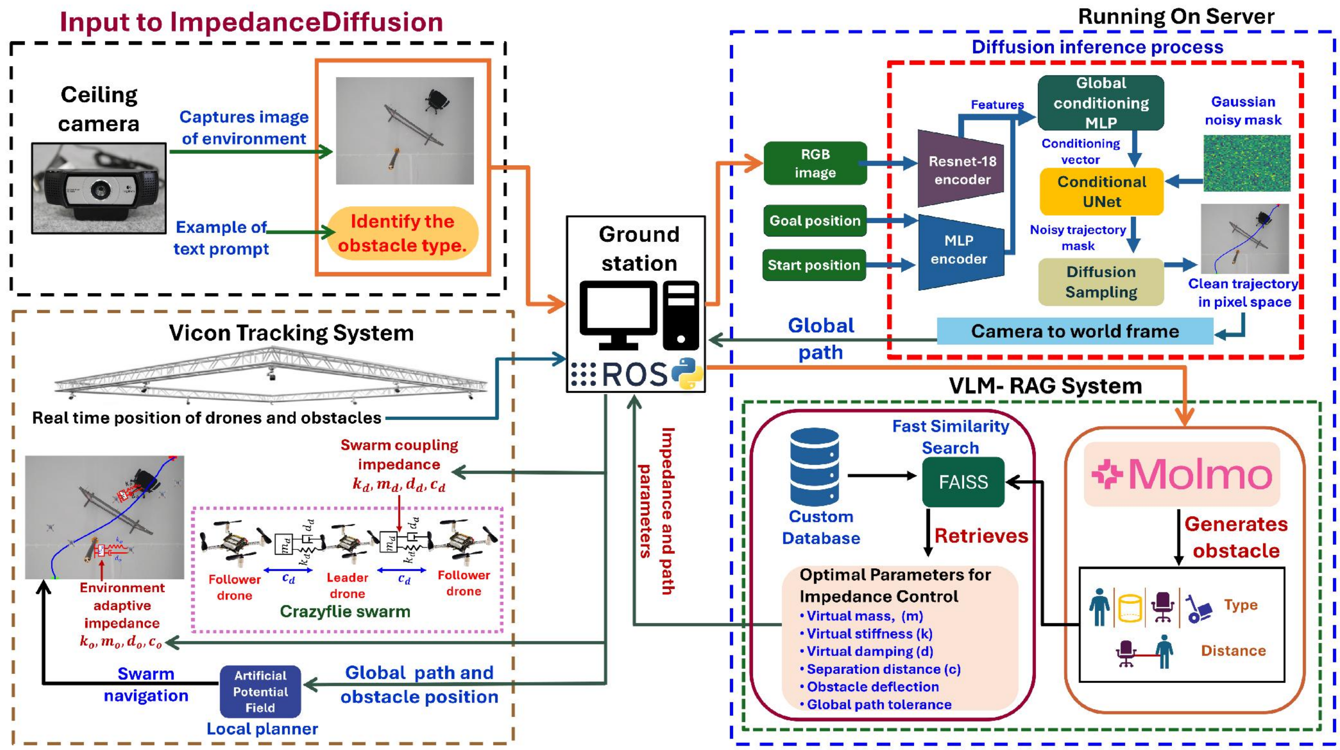

Fig. 2: System architecture of ImpedanceDiffusion. A top-view image and user prompt are processed by Molmo to identify obstacle type. Based on Molmo's output, the VLM–RAG module retrieves the corresponding impedance parameters from a custom database. In parallel, the image is provided to the diffusion model to generate a 2D global trajectory, which is executed using APF-based tracking with adaptive inter-drone and drone–obstacle impedance control. The planning is performed in 2D, while the flight altitude is maintained constant during execution.

A. Semantic Obstacle Identification via VLM–RAG

The VLM–RAG module follows the same semantic reasoning pipeline introduced in ImpedanceGPT [1], with extensions to support heterogeneous environments. A Vision-Language Model (Molmo-7B-O [21]) analyzes the top-view image to identify obstacle categories. The resulting semantic description is embedded and queried against a custom vector database using FAISS-based similarity search to retrieve impedance configurations suited to the detected scenario.

B. Custom Database for Variable Impedance Scenarios

The impedance parameter database is constructed from 200 real-world swarm flight experiments, with parameter sets selected based on formation stability, oscillation behavior, jitter reduction, and obstacle clearance performance. The chosen configurations ensure stable and collision-free navigation across repeated trials for each obstacle category. The database stores the virtual mass (m), stiffness (k), damping (d), obstacle deflection limits, global path tolerance, and separation distance (c) for both drone–drone and drone–obstacle interactions. The resulting parameter sets are summarized in Table I. At runtime, the system retrieves the appropriate parameters based on the detected obstacle class.

C. Diffusion-Based Global Planning

The global planner is a conditional UNet-based diffusion model. The model predicts a pixel-space trajectory mask by iteratively denoising a noisy trajectory sample. The input is a three-channel mask

TABLE I: EXPERIMENTALLY VALIDATED SAFE VIRTUAL IMPEDANCE LINK AND PATH PARAMETERS FOR DRONE–DRONE AND DRONE–OBSTACLE INTERACTIONS, STORED IN THE CUSTOM DATABASE FOR EACH OBSTACLE TYPE

Drone–Drone Parameters	**Cyl.**	**Chair**	**Trolley**	**Gate**	**Human**
Virtual Mass (kg)	1	1	0.8	1	5
Virtual Stiffness (N/m)	7	7	7	7	1
Virtual Damping (Ns/m)	3	3	3	3	2
Drone–Obstacle Parameters	**Cyl.**	**Chair**	**Trolley**	**Gate**	**Human**
Virtual Mass (kg)	1	0.8	0.8	1.2	1
Virtual Stiffness (N/m)	9	10	5	8	16
Virtual Damping (Ns/m)	5	5.5	3	5	4
Path Parameters	**Cyl.**	**Chair**	**Trolley**	**Gate**	**Human**
Separation Distance (m)	0.5	0.5	0.55	0.4	0.55
Obstacle Deflection (m)	0.65	0.8	1.2	0.45	1
Global Path Tolerance (m)	0.3	0.4	0.5	0.5	0.5

$$x_0 \in \mathbb{R}^{B \times 3 \times H \times W},$$

where B is the batch size, H and W are spatial dimensions, and the three channels encode the start pixel, goal pixel, and trajectory mask.

a) Forward Diffusion Process.

Noise is progressively added to the clean mask using a squared-cosine schedule:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon, \tag{1}$$

where x_t denotes the noisy trajectory mask at diffusion step t, $\alpha_t = 1 - \beta_t$ with β_t the variance schedule, $\overline{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ is the cumulative noise coefficient, and $\epsilon \sim \mathcal{N}(0, I)$ is the gaussian noise.

b) Reverse Denoising Process.

During inference, the conditional UNet predicts the clean mask $\hat{x}_0$, which is used in the Denoising Diffusion Probabilistic Model (DDPM) posterior:

$$x_{t-1} = \mu_t(x_t, \hat{x}_0) + \sigma_t z, \quad z \sim \mathcal{N}(0, I), \tag{2}$$

where x_{t-1} denotes the progressively denoised trajectory mask at step $t-1$, and μ_t and σ_t denote the posterior mean and variance. The denoising process iterates from $t = T$ to $t = 0$, while start and goal channels are inpainted at each step to enforce boundary consistency. Here, T denotes the total number of diffusion time steps.

c) Training Objective.

The training objective combines trajectory reconstruction and endpoint accuracy:

$$\mathcal{L} = \lambda_{\text{path}} L_{\text{path}} + \lambda_{\text{endpoint}} L_{\text{endpoint}}, \tag{3}$$

where λ_{path} and $\lambda_{\text{endpoint}}$ are the weights for trajectory reconstruction and endpoint accuracy respectively. The trajectory reconstruction loss is

$$L_{\text{path}} = w_t \frac{1}{N} \sum_{B,H,W} \left(T^{\text{pred}}_{B,H,W} - T^{\text{gt}}_{B,H,W} \right)^2, \tag{4}$$

where T^{pred} and T^{gt} are the predicted and ground-truth masks, respectively, w_t is the trajectory-channel weight, and and $N = B \times H \times W$ is the total number of pixels across the batch. The endpoint loss equals:

$$L_{\text{endpoint}} = \frac{1}{2} \Big[w_s \mathcal{L}_s + w_g \mathcal{L}_g \Big], \tag{5}$$

where $\mathcal{L}_s$ and $\mathcal{L}_g$ denote start and goal reconstruction losses with channel weights w_s and w_g respectively. The trajectory reconstruction loss ensures geometric fidelity, while the endpoint loss enforces boundary correctness.

1) Training Pipeline and Dataset Generation

We employ two diffusion-based planners. The top-view planner, referred to as **Diffusion Planner 1**, is designed for long-horizon global planning. The second planner, referred to as **Diffusion Planner 2**, is trained on first-person-view (FPV) or egocentric observations and is inherently suited for short-horizon planning. It is an extended version of the diffusion framework introduced in [22]. These terms are used consistently throughout the paper. Both planners are trained using trajectories generated by an A* planner, with the primary difference lying in the environments and visual modalities used for dataset generation.

For **Diffusion Planner 1**, 10,000 trajectories are generated from top-view images in the ProcTHOR environment [23], with 1,500 validation samples and 1,647 test samples. For **Diffusion Planner 2**, 13,000 trajectories are generated in simulated egocentric indoor environments from [24], divided into 10,000 training, 1,500 validation, and 1,500 test samples.

All trajectories are restricted to traversable regions, implicitly enforcing collision-free behavior. Training with 100 diffusion steps for 30 epochs achieved the best performance. The model is trained entirely in simulation and deployed in real-world experiments without fine-tuning, demonstrating zero-shot sim-to-real transfer.

D. Local APF Tracking

The diffusion model provides a global trajectory, while real-time tracking and reactive avoidance are handled by the Artificial Potential Field (APF) planner applied to the leader drone. At each control step, the next waypoint on the diffusion path is treated as a temporary goal. The total force acting on the leader is

$$F_{\text{total}} = F_{\text{attraction}} + F_{\text{repulsion}}. \tag{6}$$

The attractive force toward the waypoint is

$$F_{\text{attraction}}(d_{\text{g}}) = k_{\text{att}} d_{\text{g}}, \tag{7}$$

where d_{g} is the distance to the waypoint and k_{att} is the attraction gain.

The repulsive force from nearby obstacles is defined as

$$F_{\text{rep}}(d_{\text{obs}}) = \begin{cases} \mathbf{0}, & d_{\text{obs}} > d_{\text{safe}}, \\ k_{\text{rep}} \left(\frac{1}{d_{\text{obs}}} - \frac{1}{d_{\text{safe}}} \right) \frac{1}{d_{\text{obs}}^2}, & d_{\text{obs}} \leq d_{\text{safe}}, \end{cases} \tag{8}$$

where $d_{\text{obs}} = \|\mathbf{x} - \mathbf{o}\|$ is the Euclidean distance between the drone position $\mathbf{x}$ and the obstacle position $\mathbf{o}$, d_{safe} is the safety distance within which the obstacle exerts a repulsive influence and k_{rep} is the gain of repulsion.

E. Impedance Control for Swarm Coordination and Obstacle Interaction

To achieve compliant swarm coordination and safe obstacle avoidance, two distinct impedance mechanisms are employed: (i) a dynamic drone–drone formation impedance and (ii) a distance-based drone–obstacle interaction impedance.

1) Drone–Drone Formation Impedance

Each follower drone maintains a virtual impedance link with the leader. The leader velocity is computed as

$$\mathbf{v}_L(t) = \frac{\mathbf{x}_L(t) - \mathbf{x}_L(t - \Delta t)}{\Delta t}, \tag{9}$$

where $\mathbf{x}_L$ denotes the leader position and Δt is the sampling interval. The leader velocity and acceleration are used by the virtual interaction, which is modeled as a mass–spring–damper system. For each follower, the impedance dynamics are governed by

$$m_d \ddot{\mathbf{z}} + d_d \dot{\mathbf{z}} + k_d \mathbf{z} = F_{\text{ext}}(t), \tag{10}$$

where:

- $\mathbf{z}$ is the virtual formation displacement,
- m_d, d_d, k_d are the virtual mass, damping, and stiffness parameters for drone–drone formation,
- and $F_{\text{ext}}(t)$ represents the virtual external force induced by the leader drone.

The final follower target position x_j, in a drone formation is

$$\mathbf{x}_j = \mathbf{x}_L + \beta \mathbf{z} + \mathbf{r}_j, \tag{11}$$

where β is a scaling factor and $\mathbf{r}_j$ is the geometric formation offset defined as:

$$\mathbf{r}_j = \begin{bmatrix} R\cos\theta_j \\ R\sin\theta_j \\ 0 \end{bmatrix}, \tag{12}$$

here R is the separation distance and θ_j defines the angular distribution of followers.

The inter-drone impedance parameters (m_d, d_d, k_d) are initialized to their default values from Table I. When a follower approaches a human, these parameters are switched to the corresponding human-specific values based on the hysteresis condition:

$$\text{NearHuman}_j = \begin{cases} 1, & d_j \leq d_{\text{enter}} \\ 0, & d_j \geq d_{\text{exit}} \end{cases}, \tag{13}$$

where d_j is the minimum distance between follower drone j and a human. The parameters revert to their default values when the drone exits the interaction region.

2) Drone–Obstacle Interaction Impedance

When a follower enters a predefined deflection radius around the nearest obstacle, a temporary impedance interaction is activated. Let $\mathbf{x}_j$ be the follower position and $\mathbf{x}_o$ the obstacle position. The relative vector and distance are defined as:

$$\mathbf{r} = \mathbf{x}_j - \mathbf{x}_o, \qquad d = \|\mathbf{r}\|. \tag{14}$$

If $d < d_{\text{def}}$, where d_{def} is the deflection distance, the penetration depth is defined as:

$$\delta = d_{\text{def}} - d. \tag{15}$$

To incorporate impedance behavior, the temporal variation of the penetration depth is computed in discrete time. The penetration velocity and acceleration are approximated as:

$$\dot{\delta}_t = \frac{\delta_t - \delta_{t-1}}{\Delta t}, \qquad \ddot{\delta}_t = \frac{\dot{\delta}_t - \dot{\delta}_{t-1}}{\Delta t}, \tag{16}$$

where Δt denotes the control timestep. These quantities represent the rate at which the follower enters or exits the obstacle interaction region. The normal direction of interaction is defined as

$$\hat{\mathbf{n}} = \frac{\mathbf{r}}{\|\mathbf{r}\|}. \tag{17}$$

A virtual impedance force along the obstacle normal is then computed using a mass–spring–damper model:

$$F_n = k_o\delta + d_o\dot{\delta} + m_o\ddot{\delta}, \tag{18}$$

where k_o, d_o, and m_o are obstacle-dependent virtual stiffness, damping, and mass parameters, respectively. In practice, the acceleration term may have limited influence due to small penetration variations; however, it is retained for completeness of the impedance formulation. The parameters k_o, d_o, and m_o are selected according to the detected obstacle class from the database. Soft obstacles use lower stiffness and higher damping for compliant deflection, while rigid obstacles use higher stiffness for stronger repulsion. Gate structures are assigned the same impedance parameters to enable stable and coordinated passage through the opening. The obstacle-induced displacement is obtained by converting the virtual force into a position correction through discrete-time integration. Using a constant-acceleration approximation, the induced displacement is given by

$$\mathbf{u}_{\text{obs}} = \frac{1}{2}\frac{F_n}{m_o}\Delta t^2\,\hat{\mathbf{n}}. \tag{19}$$

The updated follower position becomes

$$\mathbf{x}_j^{\text{new}} = \mathbf{x}_j + \mathbf{u}_{\text{obs}}. \tag{20}$$

Unlike formation impedance, the obstacle interaction is implemented in discrete time and avoids continuous-time ODE integration.

IV. Experimental Setup

A total of 20 experimental configurations were evaluated across eight distinct scenario types to assess the proposed *ImpedanceDiffusion* framework in terms of global trajectory generation, semantic understanding, and variable impedance-based swarm navigation. The experiments covered static and dynamic scenarios, as well as sparse and cluttered environments, to assess robustness under varying complexity levels. Top-down images used for semantic reasoning were captured under consistent daylight conditions to ensure reliable visual perception. Multi-colored obstacles were introduced to evaluate the semantic classification capability of the VLM and mitigate potential color bias. Two diffusion models were evaluated. The top-view planner performs single-pass inference to generate the complete trajectory. The second planner, trained on FPV data, is applied using a two-stage inference pipeline, where trajectories are generated sequentially from the start to an intermediate waypoint and then to the final goal.

The VLM–RAG and diffusion models were executed on a high-performance workstation equipped with an NVIDIA RTX 4090 GPU (24GB VRAM) and an Intel Core i9-13900K processor. Due to memory constraints, the quantized Molmo-7B-O BnB 4-bit model was employed for semantic reasoning. For real-world validation, Crazyflie 2.1 drones were used to implement swarm navigation in physical indoor environments [25]. The VLM–RAG module received the following structured prompt for obstacle detection: *"Analyze the drone arena image and identify all the obstacles present on the white background of the image."*

V. Experimental Results

A total of 20 experiments were conducted across eight distinct scenarios, encompassing static and dynamic environments with hard, soft, and mixed obstacle configurations. Each experiment was repeated five times. These repeated trials were performed to evaluate the success rate of the proposed framework. Across the resulting 100 runs, the framework achieved a **92%** success rate, with failures primarily attributed to hardware limitations or communication loss. No failures were attributed to trajectory generation or impedance control instability. In this paper, we present eight representative experiments to illustrate the characteristic behaviors of the proposed framework under different environmental conditions.

A. VLM–RAG Retrieval

Across the 20 experimental configurations in 8 scenario types, the VLM–RAG module was evaluated on its ability to correctly identify the object type and retrieve the corresponding impedance parameters. Out of the 20 cases, 18 were correctly classified and matched with the appropriate impedance parameter set, resulting in a retrieval accuracy of **90%**.

B. Swarm Navigation in Static Environment

1) Experiment 1: Hard Obstacles with Gate

Fig. 3 presents a static environment containing a cylindrical obstacle, a chair, and a gate. Both diffusion planners are provided with the same top-view image, along with identical start and goal inputs. Diffusion Planner 1 generates trajectories using top-view conditioning, while Diffusion Planner 2, although trained on egocentric (FPV) data, is applied here using the same top-view input through the two-stage inference pipeline. Diffusion Planner 1 produces smoother and more direct trajectories. In contrast, Planner 2 maintains greater clearance from obstacle boundaries, which is reflected in the velocity-colored plots. When navigating near hard obstacles, the follower speed under Planner 1 remains between 1.0–1.2 m/s, while Planner 2 maintains higher speeds of 1.5–2.0 m/s.

For hard obstacles, high stiffness and moderate damping produce small deflections and reduced compliance, maintaining tight leader–follower coupling and enabling obstacle avoidance based on relative position with minimal lateral deviation. Minor oscillations arise when the follower momentarily advances ahead of the leader and is pulled back by the impedance link, appearing as small jitters in the plots.

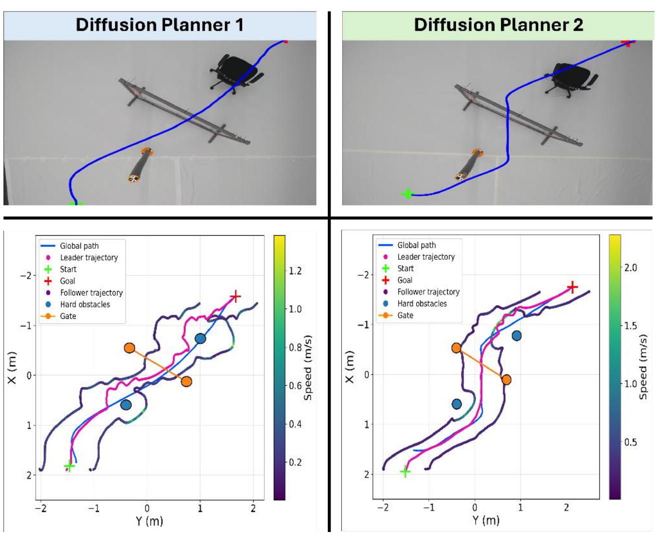

Fig. 3: Static environment with hard obstacles and a gate.

2) Experiment 2: Cluttered Scene with Hard and Soft Obstacles

Fig. 4 illustrates a cluttered static environment containing four cylindrical obstacles and one human. Around hard obstacles, both planners exhibit rigid compliance characterized by small deflection distances and relatively high speeds, approximately 1.0 m/s for Planner 1 and 1.6 m/s for Planner 2. In contrast, the response near the soft obstacle (human) differs significantly. The follower speed decreases to approximately 0.6 m/s for both planners, and the deflection distance increases compared to hard obstacles. This behavior aligns with the impedance switching strategy proposed in ImpedanceGPT [1], where humans are assigned lower stiffness and moderate damping. In our case, these parameters are slightly higher due to the use of full variable impedance control. Furthermore, Planner 2 consistently maintains higher speeds overall, as its FPV-based local diffusion planning provides greater clearance.

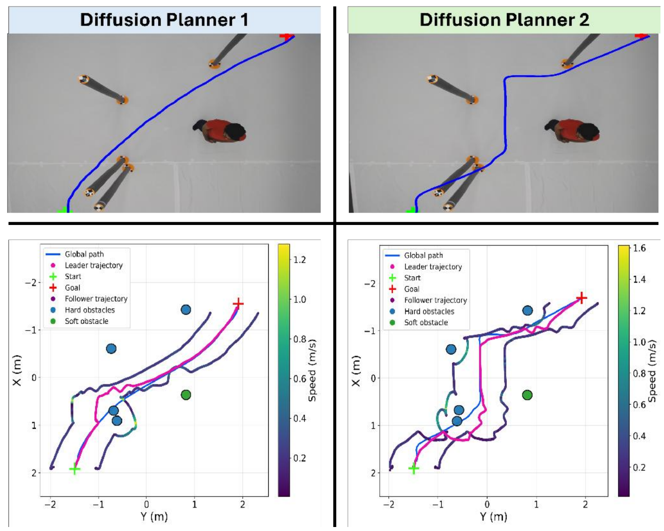

Fig. 4: Static cluttered environment with hard obstacles and one human.

C. Swarm Navigation in Dynamic Environment

1) Experiment 3: Single Dynamic Human with Hard Obstacle

Fig. 5 presents a dynamic environment containing a moving human and a static chair. The planners exhibit distinct behaviors. Planner 1 generates trajectories that pass closer to obstacles, whereas Planner 2 maintains larger local clearance due to FPV-based conditioning, leading to different velocity profiles. Near the human, Planner 1 reduces speed to approximately 0.8 m/s, while Planner 2 maintains a higher speed of about 1.6 m/s. Higher velocity near a moving human reduces reaction margin and may increase risk in rapidly changing scenarios. Rigid compliance is observed around the chair, characterized by small separation distances. In contrast, soft compliance near the human results in larger path deflections. Because Planner 1 operates closer to the human, it experiences stronger APF corrections and sharper local velocity variations.

2) Experiment 4: Two Dynamic Soft Obstacles

Fig. 6 presents a fully dynamic scenario with two moving humans. Both planners exhibit soft compliant behavior, maintaining large deflection distances around the humans. During avoidance, the follower speed is approximately 1.0 m/s for Planner 1 and 1.6 m/s for Planner 2. Planner 2 maintains higher speeds due to its FPV-based local condi-

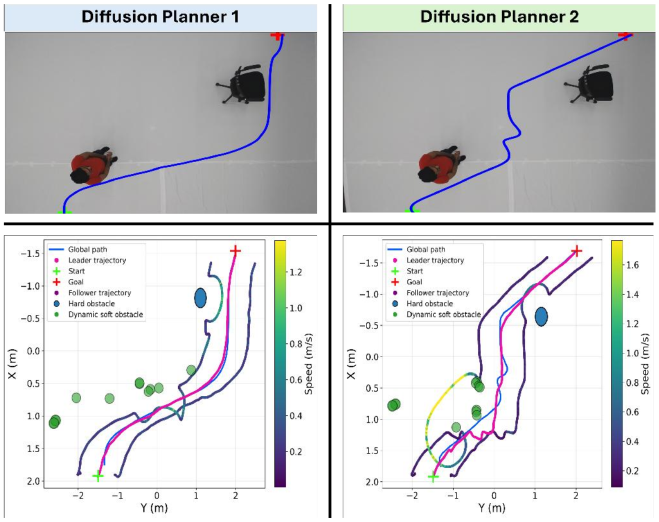

Fig. 5: Dynamic environment with one moving human and one hard obstacle.

tioning, which produces trajectories with greater immediate clearance and consequently smaller APF-induced repulsive forces. As a result, fewer corrective actions are required compared to Planner 1. Minor trajectory overlaps and small jitters appear due to impedance-based leader–follower coupling. The oscillations remain bounded and decay over time, confirming stable closed-loop impedance behavior under dynamic conditions.

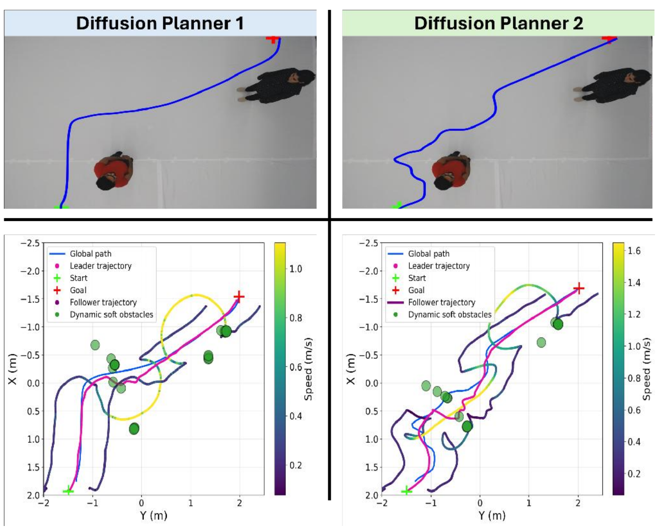

Fig. 6: Dynamic scenario with two humans.

D. Comparative Analysis of Diffusion Planners

We compare two diffusion-based trajectory planners trained under different visual modalities. **Diffusion Planner 1 (P1)** trained on top-view images and requires a single inference requiring 1.4–2.5 s per trajectory. **Diffusion Planner 2 (P2)** is trained on FPV images and is deployed on top-view inputs in zero shot manner using a two-stage inference pipeline, resulting in a longer inference time of 2.5–3.4 s per trajectory. Both planners achieved a **100% trajectory generation rate** across all eight experimental scenarios. Since no trajectory generation failures were observed, the comparison focuses on trajectory quality metrics rather than binary success.

TABLE II: PERFORMANCE COMPARISON ACROSS EIGHT EXPERIMENTAL SCENARIOS

Exp.	Scenario	Planner	Path (m)	Coll. (-)	Goal (m)	Turn (rad)
E1	Cyl.–Gate	P1	2.013	0.186	0.026	10.541
		P2	2.191	0.067	0.028	7.279
E2	2 Cyl.–Gate	P1	2.148	0.352	0.044	9.254
		P2	2.235	0.146	0.047	9.022
E3	Cyl.–Gate–Chair	P1	2.093	0.338	0.016	11.333
		P2	2.270	0.369	0.147	8.553
E4	Chair–Trolley	P1	2.198	0.486	0.050	9.811
		P2	2.191	0.295	0.050	10.216
E5	5 Cyl.	P1	2.105	0.693	0.044	9.190
		P2	2.351	0.447	0.045	11.812
E6	4 Cyl.–Human	P1	2.078	0.380	0.044	9.893
		P2	2.366	0.447	0.045	9.304
E7	Human–Chair	P1	2.338	0.318	0.034	8.099
		P2	2.354	0.119	0.034	15.313
E8	2 Human	P1	2.339	0.036	0.045	7.117
		P2	2.509	0.085	0.047	16.760

The evaluation metrics are:

- **Path Length (m):** Total arc length of the trajectory.
- **Collision Ratio (-):** Fraction of trajectory samples intersecting inflated obstacle regions. High collision ratios correspond to overlap with inflated pixel-space safety regions and do not indicate physical collisions during real-world execution.
- **Goal Error (m):** Euclidean distance between the final trajectory point and the ground-truth goal.
- **Total Turning (rad):** Cumulative absolute heading change along the trajectory. Lower cumulative turning suggests smoother and more dynamically efficient trajectories.

Across all scenarios, both planners produce comparable path lengths, confirming that the FPV-based planner can effectively perform global planning when required. The average path length difference between P1 and P2 remains marginal across experiments.

Planner 2 demonstrates a lower average collision ratio of (**0.246**) compared to Planner 1 (**0.348**), indicating improved obstacle clearance despite its cross-modal deployment, while goal accuracy remains comparable across all scenarios. In terms of geometric complexity, Planner 1 exhibits lower average cumulative turning (**9.41 rad** vs. **11.03 rad**), suggesting more directionally efficient trajectories, whereas the increased turning in Planner 2 is consistent with its iterative FPV-based inference strategy. Overall, these results indicate that Planner 1 offers faster inference and slightly smoother paths, while Planner 2 achieves competitive global performance with stronger clearance behavior. Across all experiments, the framework validates online impedance switching based on obstacle type, where hard obstacles induce small deflections due to higher stiffness and humans result in softer, larger compliant deviations. Moreover, impedance coupling preserves stable formation with only minor bounded oscillations and no collisions.

VI. CONCLUSION AND FUTURE WORK

This paper introduced *ImpedanceDiffusion*, a hierarchical swarm navigation framework that leverages image-conditioned diffusion-based global planning, APF-based reactive tracking, and semantic-aware variable impedance control. The proposed system enables map-free navigation of small aerial drone swarm in mixed indoor environments with soft and hard obstacles without explicit geometric mapping or classical search-based planners.

Across 20 experimental configurations (100 total flight runs), the framework achieved a **92% success rate**, with failures primarily attributed to hardware or communication limitations rather than planning instability. Both diffusion planners demonstrated a **100% trajectory generation rate** across all evaluated scenarios, confirming the reliability of diffusion-based global planning under diverse environmental conditions. The VLM–RAG module achieved a **90% retrieval accuracy** in identifying obstacle classes and selecting corresponding impedance parameters. Quantitatively, the FPV-conditioned planner exhibited stronger obstacle clearance behavior (lower average collision ratio of 0.246 compared to 0.348 for the top-view planner), while the top-view planner produced smoother trajectories with lower cumulative turning and faster inference. These results highlight a clear trade-off between geometric efficiency and local clearance robustness depending on visual conditioning modality. Overall, the findings demonstrate that diffusion models can serve as reliable global planners for image-conditioned swarm navigation, while semantic impedance adaptation enables obstacle-specific compliant behavior without compromising closed-loop stability. The integration of generative planning with physically grounded impedance control provides a practical pathway toward scalable, map-free, and semantically aware swarm navigation.

Future work will focus on enabling onboard perception and decentralized inference to remove reliance on workstation-level computation. Additionally, learning continuous impedance adaptation policies rather than discrete class-based switching will further improve scalability in dense, human-shared environments.

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